6. Resets

6.1 Overview

There are nine types of resets: RES# pin reset, power-on reset, voltage-monitoring 0 reset, voltage-monitoring 1 reset, voltage-monitoring 2 reset, deep software standby reset, independent watchdog timer reset, watchdog timer reset, and software reset.

Table 6.1 lists the reset names and sources.

Table 6.1 Reset Names and Sources

Reset Name	Source
RES# pin reset	Voltage input to the RES# pin is driven low.
Power-on reset	VCC rises (voltage detection: VPOR)*1
Voltage-monitoring 0 reset	VCC falls (voltage detection: Vdet0)*1
Voltage-monitoring 1 reset	VCC falls (voltage detection: Vdet1)*1
Voltage-monitoring 2 reset	VCC falls (voltage detection: Vdet2)*1
Deep software standby reset	Deep software standby mode is canceled by an interrupt.
Independent watchdog timer reset	The independent watchdog timer underflows, or a refresh error occurs.
Watchdog timer reset	The watchdog timer underflows, or a refresh error occurs.
Software reset	Register setting

Note 1. For details on the voltages to be monitored (VPOR, Vdet0, Vdet1, and Vdet2), refer to section 8, Voltage Detection Circuit (LVDA) and section 63, Electrical Characteristics.

The internal state and pins are initialized by a reset.
Table 6.2 lists the reset targets to be initialized.

Table 6.2 Targets to be Initialized by Each Reset Source

Targets to be Initialized	Reset Source								
	RES# Pin Reset	Power-On Reset	Voltage-Monitoring 0 Reset	Independent Watchdog Timer Reset	Watchdog Timer Reset	Voltage-Monitoring 1 Reset	Voltage-Monitoring 2 Reset	Deep Software Standby Reset	Software Reset
Power-on reset detect flag (RSTSR0.PORF)	✓	—	—	—	—	—	—	—	—
Cold start/warm start determination flag (RSTSR1.CWSF)	—	✓	—	—	—	—	—	—	—
Voltage-monitoring 0 reset detect flag (RSTSR0.LVD0RF)	✓	✓	—	—	—	—	—	—	—
Independent watchdog timer reset detect flag (RSTSR2.IWDTRF)	✓	✓	✓	—	—	—	—	✓	—
Independent watchdog timer (IWDTRR, IWDTCR, IWDTSR, IWDTRCR, IWDTCSTPR, ILOCOCR)	✓	✓	✓	—	—	—	—	✓	—
Watchdog timer reset detect flag (RSTSR2.WDTRF)	✓	✓	✓	✓	—	—	—	✓	—
Registers related to the watchdog timer (WDTRR, WDTCR, WDTSR, WDTRCR)	✓	✓	✓	✓	—	—	—	✓	—
Voltage-monitoring 1 reset detect flag (RSTSR0.LVD1RF)	✓	✓	✓	✓	✓	—	—	—	—
Registers related to the voltage monitor function 1 (LVD1CR0, LVCMPCR.LVD1E, LVDLVLR.LVD1LVL[3:0])	✓	✓	✓	✓	✓	—	—	—	—
(LVD1CR1, LVD1SR)	✓	✓	✓	✓	✓	—	—	✓	—
Voltage-monitoring 2 reset detect flag (RSTSR0.LVD2RF)	✓	✓	✓	✓	✓	✓	—	—	—
Registers related to the voltage monitor function 2 (LVD2CR0, LVCMPCR.LVD2E, LVDLVLR.LVD2LVL[3:0])	✓	✓	✓	✓	✓	✓	—	—	—
(LVD2CR1, LVD2SR)	✓	✓	✓	✓	✓	✓	—	✓	—
Deep software standby reset detect flag (RSTSR0.DPSRSTF)	✓	✓	✓	✓	✓	✓	✓	—	—
Software reset detect flag (RSTSR2.SWRF)	✓	✓	✓	✓	✓	✓	✓	✓	—
Register related to the realtime clock*1	—	—	—	—	—	—	—	—	—
Register related to high-speed on-chip oscillator (HOCOPCR.HOCOPCNT)	✓	✓	✓	✓	✓	✓	✓	—	✓
Register related to main clock oscillator (MOFCR)	✓	✓	✓	✓	✓	✓	✓	—	✓
Pin state	✓	✓	✓	✓	✓	✓	✓	—	✓
Registers related to the low power-consumption function (DPSBYCR, DPSIER0 to DPSIER3, DPSIFR0 to DPSIFR3, DPSIEGR0 to DPSIEGR3, DPUSR0R, DPUSR1R)*2	✓	✓	✓	✓	✓	✓	✓	—	✓
Registers other than the above, CPU, and internal state	✓	✓	✓	✓	✓	✓	✓	✓	✓

✓: Targets to be initialized, —: No change occurs.

Note 1. Some control bits are initialized by all types of resets. For details on the target bits, refer to section 33, Realtime Clock (RTCd).

Note 2. Of the registers related to the low-power-consumption function, the DPSBKRy register is not initialized by any reset. For details, refer to section 11, Low Power Consumption.

When a reset is canceled, the reset exception handling starts. For details on the reset exception handling, refer to section 14, Exception Handling.

Table 6.3 lists the pin related to the reset.

Table 6.3 Pin Related to Reset

Pin Name	I/O	Function
RES#	Input	Reset pin

6.2 Register Descriptions

6.2.1 Reset Status Register 0 (RSTSR0)

Address(es): 0008 C290h

	b7	b6	b5	b4	b3	b2	b1	b0
	DPSRSTF	—	—	—	LVD2RF	LVD1RF	LVD0RF	PORF
Value after reset:	0*1	0	0	0	0*1	0*1	0*1	0*1

Bit	Symbol	Bit Name	Description	R/W
b0	PORF	Power-On Reset Detect Flag	0: Power-on reset not detected. 1: Power-on reset detected.	R(/W) *2
b1	LVD0RF	Voltage-Monitoring 0 Reset Detect Flag	0: Voltage-monitoring 0 reset not detected. 1: Voltage-monitoring 0 reset detected.	R(/W) *2
b2	LVD1RF	Voltage-Monitoring 1 Reset Detect Flag	0: Voltage-monitoring 1 reset not detected. 1: Voltage-monitoring 1 reset detected.	R(/W) *2
b3	LVD2RF	Voltage-Monitoring 2 Reset Detect Flag	0: Voltage-monitoring 2 reset not detected. 1: Voltage-monitoring 2 reset detected.	R(/W) *2
b6 to b4	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b7	DPSRSTF	Deep Software Standby Reset Flag	0: Deep software standby mode cancelation not requested by an interrupt. 1: Deep software standby mode cancelation requested by an interrupt.	R(/W) *2

Note 1. The value after reset depends on the reset source.

Note 2. Only 0 can be written to clear the flag.

PORF Flag (Power-On Reset Detect Flag)

The PORF flag indicates that a power-on reset has occurred.

[Setting condition]

- When a power-on reset occurs.

[Clearing conditions]

- When a reset listed in Table 6.2 occurs.
- When PORF is read as 1 and then 0 is written to PORF.

LVD0RF Flag (Voltage-Monitoring 0 Reset Detect Flag)

The LVD0RF flag indicates that a voltage-monitoring 0 reset has occurred due to the VCC voltage falling below Vdet0.

[Setting condition]

- When a voltage-monitoring 0 reset occurs.

[Clearing conditions]

- When a reset listed in Table 6.2 occurs.
- When LVD0RF is read as 1 and then 0 is written to LVD0RF.

LVD1RF Flag (Voltage-Monitoring 1 Reset Detect Flag)

The LVD1RF flag indicates that a voltage-monitoring 1 reset has occurred due to the VCC voltage falling below Vdet1.

[Setting condition]

- When a voltage-monitoring 1 reset occurs.

[Clearing conditions]

- When a reset listed in Table 6.2 occurs.
- When LVD1RF is read as 1 and then 0 is written to LVD1RF.

LVD2RF Flag (Voltage-Monitoring 2 Reset Detect Flag)

The LVD2RF flag indicates that a voltage-monitoring 2 reset has occurred due to the VCC voltage falling below Vdet2.

[Setting condition]

- When a voltage-monitoring 2 reset occurs.

[Clearing conditions]

- When a reset listed in Table 6.2 occurs.
- When LVD2RF is read as 1 and then 0 is written to LVD2RF.

DPSRSTF Flag (Deep Software Standby Reset Flag)

The DPSRSTF flag indicates that deep software standby mode has been canceled by an interrupt and that an internal reset (deep software standby reset) occurred.

[Setting condition]

- When deep software standby mode is cancelled by an interrupt.
  For details, refer to section 11, Low Power Consumption.

[Clearing conditions]

- When a reset listed in Table 6.2 occurs.
- When DPSRSTF is read as 1 and then 0 is written to DPSRSTF.

6.2.2 Reset Status Register 1 (RSTSR1)

Address(es): 0008 C291h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	CWSF
Value after reset:	0	0	0	0	0	0	0	0/1*1

Bit	Symbol	Bit Name	Description	R/W
b0	CWSF	Cold/Warm Start Determination Flag	0: Cold start 1: Warm start	R(/W)*2
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. The value after reset depends on the reset source.
Note 2. Only 1 can be written to set the flag.

RSTSR1 determines whether a power-on reset has caused the reset processing (cold start) or a reset signal input during operation has caused the reset processing (warm start).

CWSF Flag (Cold/Warm Start Determination Flag)

The CWSF flag indicates the type of reset processing: cold start or warm start.

The CWSF flag is initialized by a power-on reset. It is not initialized by a reset signal generated by the RES# pin.

[Setting condition]

- When 1 is written through programming; it is not set to 0 even when 0 is written.

[Clearing condition]

- When a reset listed in Table 6.2 occurs.

6.2.3 Reset Status Register 2 (RSTSR2)

Address(es): 0008 00C0h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	SWRF	WDTRF	IWDTRF
Value after reset:	0	0	0	0	0	0*1	0*1	0*1

Bit	Symbol	Bit Name	Description	R/W
b0	IWDTRF	Independent Watchdog Timer Reset Detect Flag	0: Independent watchdog timer reset not detected. 1: Independent watchdog timer reset detected.	R(/W) *2
b1	WDTRF	Watchdog Timer Reset Detect Flag	0: Watchdog timer reset not detected. 1: Watchdog timer reset detected.	R(/W) *2
b2	SWRF	Software Reset Detect Flag	0: Software reset not detected. 1: Software reset detected.	R(/W) *2
b7 to b3	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. The value after reset depends on the reset source.
Note 2. Only 0 can be written to clear the flag.

IWDTRF Flag (Independent Watchdog Timer Reset Detect Flag)

The IWDTRF flag indicates that an independent watchdog timer reset has occurred.

[Setting condition]

- When an independent watchdog timer reset occurs.

[Clearing conditions]

- When a reset listed in Table 6.2 occurs.
- When IWDTRF is read as 1 and then 0 is written to IWDTRF.

WDTRF Flag (Watchdog Timer Reset Detect Flag)

The WDTRF flag indicates that a watchdog timer reset has occurred.

[Setting condition]

- When a watchdog timer reset occurs.

[Clearing conditions]

- When a reset listed in Table 6.2 occurs.
- When WDTRF is read as 1 and then 0 is written to WDTRF.

SWRF Flag (Software Reset Detect Flag)

The SWRF flag indicates that a software reset has occurred.

[Setting condition]

- When a software reset occurs.

[Clearing conditions]

- When a reset listed in Table 6.2 occurs.
- When SWRF is read as 1 and then 0 is written to SWRF.

6.2.4 Software Reset Register (SWRR)

Address(es): 0008 00C2h

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	SWRR[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b15 to b0	SWRR[15:0]	Software Reset	Writing A501h resets the MCU. These bits are read as 0000h.	R/W

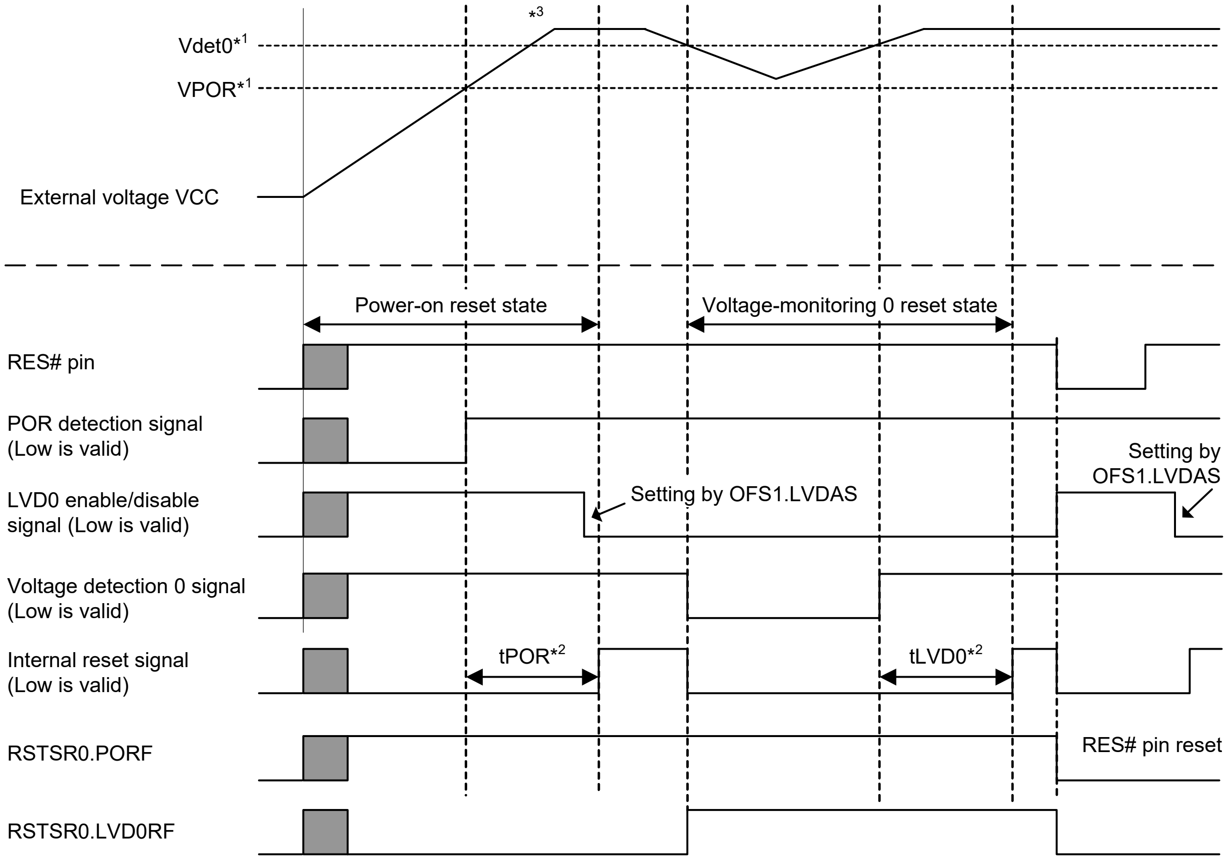
6.3 Operation

6.3.1 RES# Pin Reset

This is a reset generated by the RES# pin.
When the RES# pin is driven low, all the processing in progress is aborted and the MCU enters a reset state.
In order to unfailingly reset the MCU, the RES# pin should be held low for the specified power supply stabilization time at a power-on.
When the RES# pin is driven high from low, the internal reset is canceled after the post-RES# cancelation wait time (tRESWT) has elapsed, and then the CPU starts the reset exception handling.
For details, refer to section 63, Electrical Characteristics.

6.3.2 Power-On Reset and Voltage Monitoring 0 Reset

The power-on reset is an internal reset generated by the power-on reset circuit.
If the RES# pin is in a high level state when power is supplied, a power-on reset is generated. In addition, if the RES# pin is in a high level state when power falls (including the case when VCC falls below VPOR), a power-on reset is generated.
After VCC has exceeded VPOR and the specified period (power-on reset time) has elapsed, the internal reset is canceled and the CPU starts the reset exception handling. The power-on reset time is used for the stabilization of the power supply and the MCU circuit.
After a power-on reset has been generated, the PORF flag in the RSTSR0 is set to 1. The PORF flag is initialized by the RES# pin reset.
The voltage monitoring 0 reset is an internal reset generated by the voltage monitoring circuit. If the voltage detection 0 circuit start (LVDAS) bit in the option function select register 1 (OFS1) is 0 (voltage monitoring 0 reset is enabled after a reset) and VCC falls below Vdet0, the RSTSR0.LVD0RF flag becomes 1 and the voltage detection circuit generates voltage monitoring 0 reset. Clear the OFS1.LVDAS bit to 0 if the voltage monitoring 0 reset is to be used.
After VCC has exceeded Vdet0 and the voltage-monitoring 0 reset time (tLVD0) has elapsed, the internal reset is canceled and the CPU starts the reset exception handling. The Vdet0 voltage detection level can be changed by the setting of the VDSEL[1:0] bits in the option function select register 1 (OFS1).
Figure 6.1 shows operations during a power-on reset and voltage monitoring 0 reset.
For details on voltage monitoring 0 reset, refer to section 8, Voltage Detection Circuit (LVDA).



Note: For details on the electrical characteristics, refer to section 63, Electrical Characteristics.
Note 1. Vdet0 shows a voltage monitoring 0 reset detection level, and VPOR shows a power-on reset detection level.
Note 2. tPOR shows a time for power-on reset, and tLVD0 shows a time for voltage monitoring 0 reset.
Note 3. At a power-on, VCC should be risen to the minimum guaranteed voltage before the POR reset is released.

Figure 6.1 Operation Examples During a Power-On Reset and Voltage Monitoring 0 Reset

6.3.3 Voltage Monitoring 1 Reset and Voltage Monitoring 2 Reset

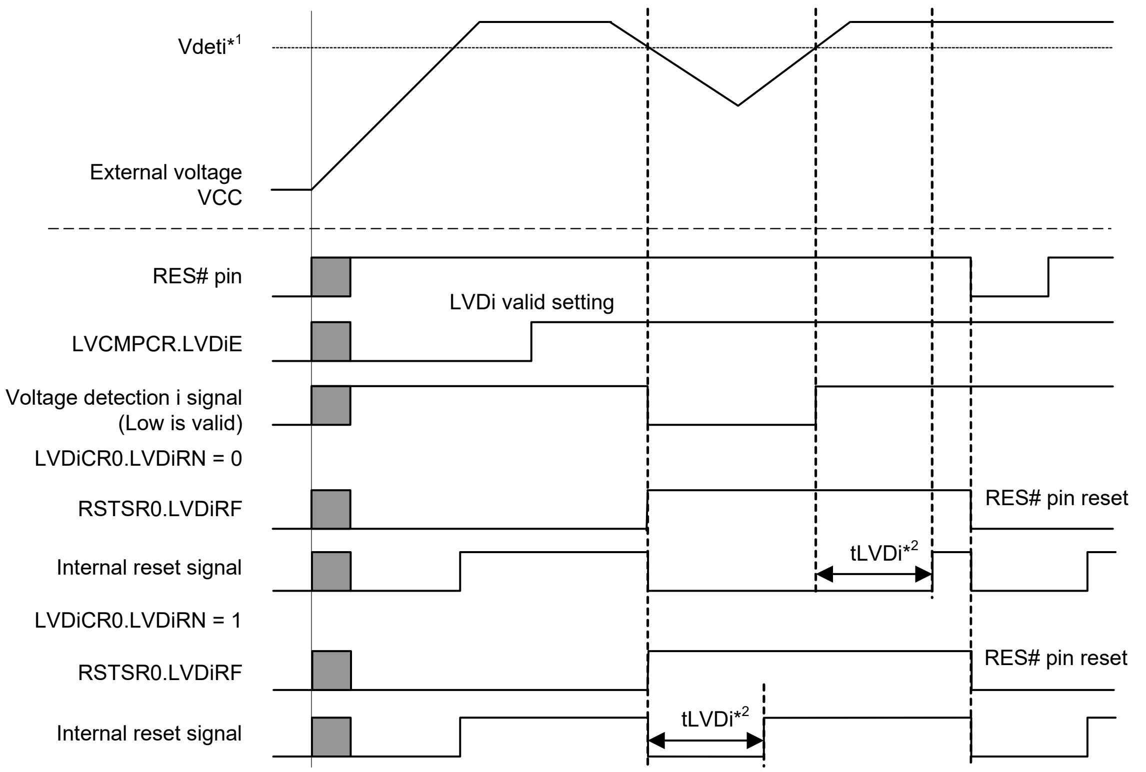
The voltage monitoring 1 reset and voltage monitoring 2 reset are internal resets generated by the voltage monitoring circuit.

When the voltage monitoring 1 interrupt/reset enable bit (LVD1RIE) is set to 1 (enabling generation of a reset or interrupt by the voltage detection circuit) and the voltage monitoring 1 circuit mode select bit (LVD1RI) is set to 1 (selecting generation of a reset in response to detection of a low voltage) in the voltage monitoring 1 circuit control register 0 (LVD1CR0), the RSTSR0.LVD1RF flag is set to 1 and the voltage-detection circuit generates a voltage monitoring 1 reset if VCC falls to or below Vdet1.

Likewise, when the voltage monitoring 2 interrupt/reset enable bit (LVD2RIE) is set to 1 (enabling generation of a reset or interrupt by the voltage detection circuit) and the voltage monitoring 2 circuit mode select bit (LVD2RI) is set to 1 (selecting generation of a reset in response to detection of a low voltage) in voltage monitoring 2 circuit control register 0 (LVD2CR0), the RSTSR0.LVD2RF flag is set to 1 and the voltage detection circuit generates a voltage monitoring 2 reset if VCC falls to or below Vdet2.

Timing for release from the voltage monitoring 1 reset state is selectable with the voltage monitoring 1 reset negate select bit (LVD1RN) in the LVD1CR0. When the LVD1CR0.LVD1RN bit is 0 and VCC has fallen to or below Vdet1, the CPU is released from the internal reset state and starts reset exception handling once the voltage-monitoring 1 reset time (tLVD1) has elapsed after VCC has risen above Vdet1. When the LVD1CR0.LVD1RN bit is 1 and VCC has fallen to or below Vdet1, the CPU is released from the internal reset state and starts reset exception handling once the voltage-

monitoring 1 reset time (tLVD1) has elapsed.
Likewise, timing for release from the voltage monitoring 2 reset state is selectable by setting the voltage monitoring 2 reset negate select bit (LVD2RN) in LVD2CR0 register.
Detection levels Vdet1 and Vdet2 can be changed by settings in the voltage detection select register (LVDLVLR).
Figure 6.2 shows examples of operations during voltage monitoring 1 and 2 resets.
For details on the voltage monitoring 1 reset and voltage monitoring 2 reset, refer to section 8, Voltage Detection Circuit (LVDA).



Note: For details on the electrical characteristics, refer to section 63, Electrical Characteristics.
Note 1. Vdeti shows a detection level of voltage monitoring 1 reset and voltage monitoring 2 reset. (i = 1, 2)
Note 2. tLVDi shows a time for voltage monitoring 1 reset and voltage monitoring 2 reset. (i = 1, 2)

Figure 6.2 Operation Examples During Voltage Monitoring 1 Reset and Voltage Monitoring 2 Reset

6.3.4 Deep Software Standby Reset

This is an internal reset generated when deep software standby mode is released by an interrupt.
When an interrupt for releasing from deep software standby mode is generated, a deep software standby reset is generated. The deep software standby reset is negated after recovery time from deep software standby mode (tDSBY) has elapsed. At the same time, deep software standby mode is also released.
When the wait time after recovery from deep software standby mode (tDSBYWT) has elapsed after deep software standby mode has been released, the internal reset is negated and the CPU starts the reset exception handling.
For details of the deep software standby reset, refer to section 11, Low Power Consumption.

6.3.5 Independent Watchdog Timer Reset

Independent watchdog timer reset is an internal reset generated by the independent watchdog timer.
Output of the independent watchdog timer reset from the independent watchdog timer can be selected by settings in the IWDT reset control register (IWDTRCR) or option function select register 0 (OFS0).
When output of the independent watchdog timer reset is selected, an independent watchdog timer reset is generated if the independent watchdog timer underflows, or if data is written when refresh operation is disabled. When the internal reset time (tRESW2) has elapsed after the independent watchdog timer reset has been generated, the internal reset is canceled and the CPU starts the reset exception handling.
For details on the independent watchdog timer reset, refer to section 35, Independent Watchdog Timer (IWDTa).

6.3.6 Watchdog Timer Reset

The watchdog-timer reset is an internal reset from the watchdog timer.
Output of the independent watchdog timer reset from the independent watchdog timer can be selected by settings in the IWDT reset control register (IWDTRCR) or option function select register 0 (OFS0).
When output of the watchdog timer reset is selected, a watchdog timer reset is generated if the watchdog timer underflows, or if data is written when refresh operation is disabled. When the internal reset time (tRESW2) has elapsed after the watchdog timer reset is generated, the internal reset is canceled and the CPU starts the reset exception handling.
For details on the watchdog timer reset, refer to section 34, Watchdog Timer (WDTA).

6.3.7 Software Reset

The software reset is an internal reset generated by the software reset circuit.
When A501h is written to SWRR, a software reset is generated. When the internal reset time (tRESW2) has elapsed after the software reset is generated, the internal reset is canceled and the CPU starts the reset exception handling.

6.3.8 Determination of Cold/Warm Start

By reading the CWSF flag in RSTSR1, the type of reset processing caused can be identified; that is, whether a power-on reset has caused the reset processing (cold start) or a reset signal input during operation has caused the reset processing (warm start).

The CWSF flag in RSTSR1 is set to 0 when a power-on reset occurs (cold start); otherwise the flag is not set to 0. The flag is set to 1 when 1 is written to it through programming; it is not set to 0 even when 0 is written.

Figure 6.3 shows an example of cold/warm start determination operation.

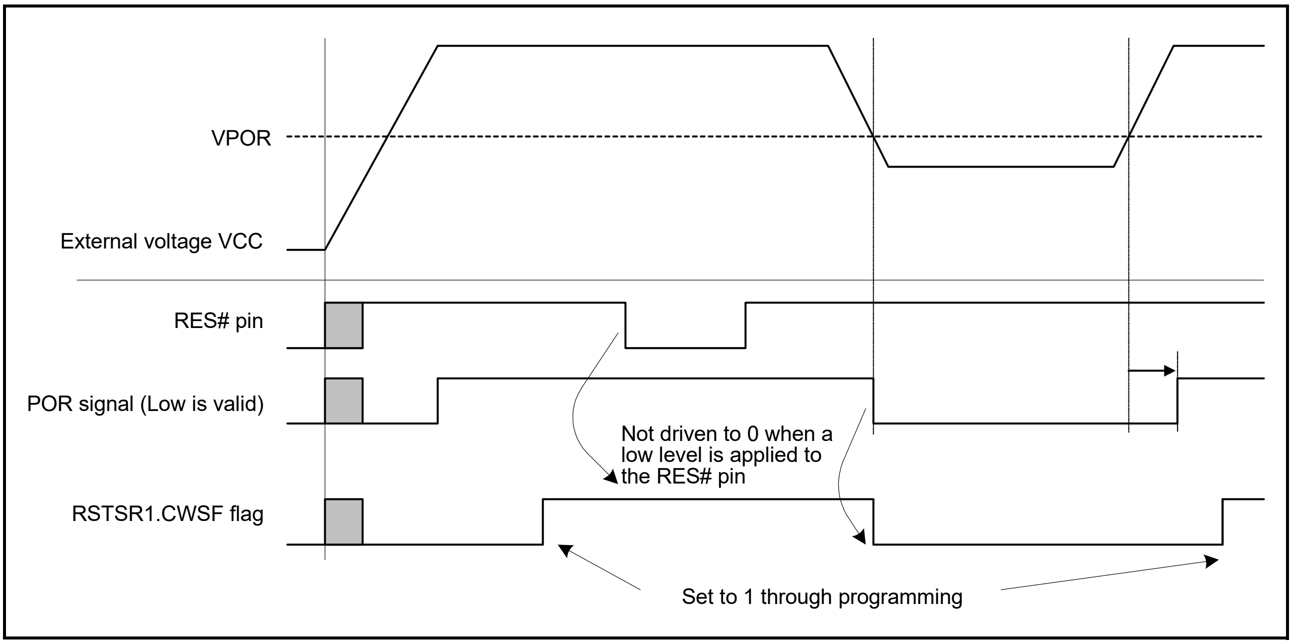


Figure 6.3 Example of Cold/Warm Start Determination Operation

6.3.9 Determination of Reset Generation Source

Reading RSTSR0 and RSTSR2 determines which reset was used to execute the reset exception handling.

Figure 6.4 shows an example of the flow to identify a reset generation source.

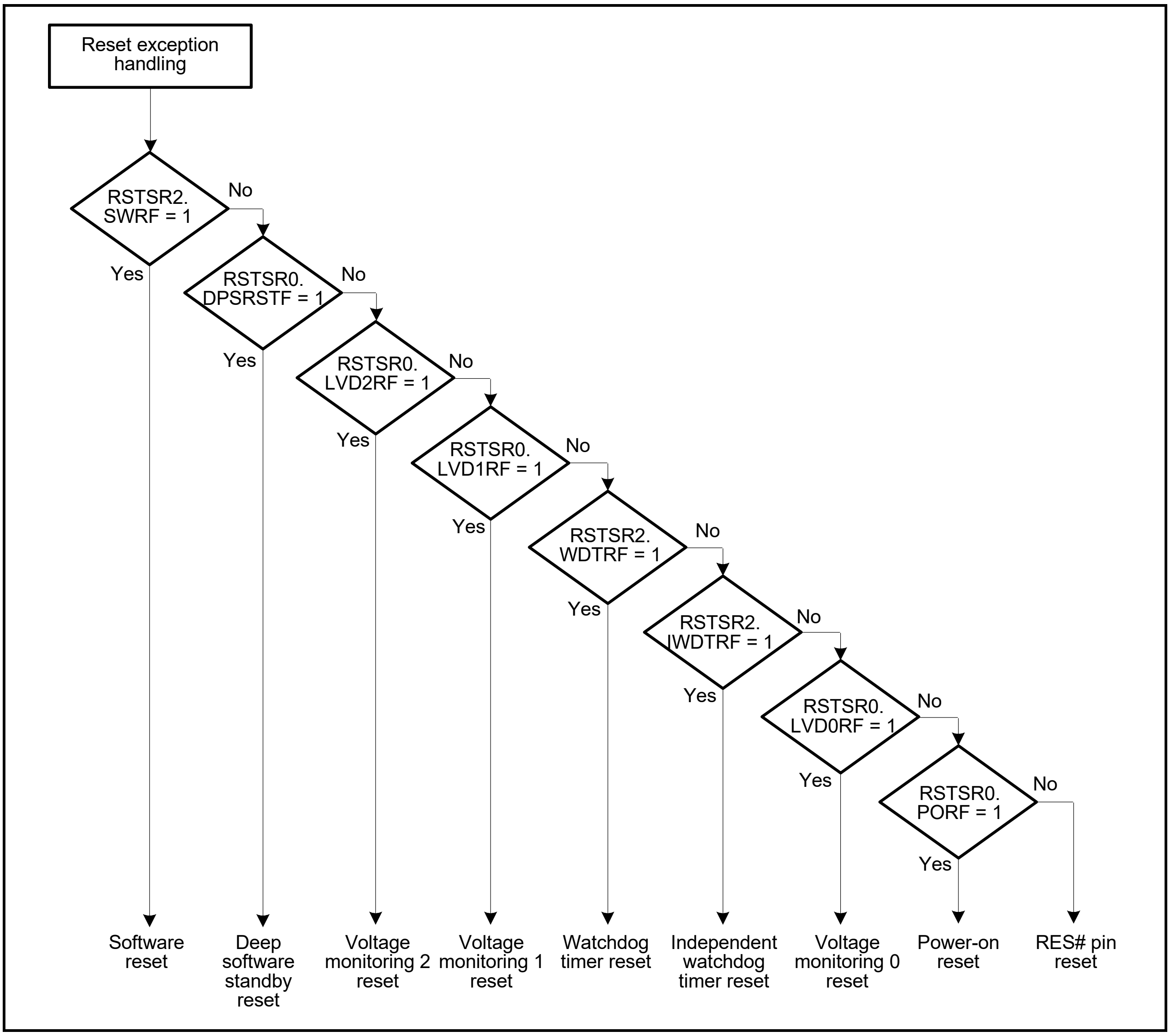


Figure 6.4 Example of Reset Generation Source Determination Flow